

5.4. Orthonormal Bases and Particulate, Integral, and Prime Wave Numbers

One may extend the n orthogonal bases $\mathbf{u}(\frac{1}{n}, j)$, $j = 1, n$ of additive wave numbers defined in Section 4.3 to bases of invertible wave numbers that are analogous to the standard bases of Euclidean geometry.

Theorem 5.24. *For every $n \in \mathbb{N}^+$, there exist n invertible wave numbers*

$$\left\{ \mathbf{e}(\frac{1}{n}, \xi) = (0, \dots, 0, 1, 0, \dots, 0) \mid \xi = 1, n \right\}, \quad (93)$$

in which the 1 appears in the ξ th place, that form an orthonormal basis for representing an invertible wave number $\mathbf{Aw}(f, g)$.

Proof Using Equation 67 define

$$\mathbf{e}(\frac{1}{n}, j) = \frac{1}{n} \mathbf{u}(\frac{1}{n}, j), j = 1, n \quad (94)$$

from which it follows that the j th term of the principal sequence is

$${}_1\mathbf{e}(\frac{1}{n}, j) = (0, \dots, 1, \dots, 0) \quad (95)$$

with the 1 occurring in the j th position. Clearly the n numbers form an orthonormal set of invertible wave number under $\otimes$ and it follows from Definition 5.1 for the invertible wave numbers that the expression $\bigoplus_{j \in S} \mathbf{e}(\frac{1}{n}, j)(\mathbf{Aw}(f, g))$,

in which $\mathbf{Aw}(f, g)$ an invertible wave number of period n and S is a subset of the first n natural numbers, is an invertible wave number of period n with $\bigoplus_{j=1}^n \mathbf{Aw}(f, g)\mathbf{e}(\frac{1}{n}, \xi) = \mathbf{Aw}(f, g)$. $\square$

The wave numbers of the basis $\{\mathbf{e}(\frac{1}{n}, j), j = 1, n\}$ may be viewed as operators under $\otimes$ that select the phases that comprise a wave number. One may therefore employ these operators in constructing wave numbers that contain arbitrary subsets of the phases of an invertible wave number $\mathbf{Aw}(f, g)$ and zeros in place of the non-included phases. Three interesting examples of such classes are the integral, particulate, and prime wave numbers.

5.4.1. The Integral Wave Numbers

Integral wave numbers are sequences of numbers whose phases are the cumulative sums of the phases of a wave number. The integral wave numbers may be constructed by applying the orthonormal basis to previously-defined wave numbers in a manner specified in